

Milestone Four

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Problem Statement/Description

This paper develops a unified framework for financial crime detection utilizing account and transaction features. Applied across the IBM Transactions for AML, Paycheck Protection Program (PPP), and the European Credit Card (ECC) Fraud Detection datasets, the framework identifies common challenges like class imbalance, high false-positive rates, and overfitting. The presented solutions provide data scientists with strategies to combat financial crime in their organizations to save money, ensure regulatory compliance, and maintain customer confidence.

Exploratory Data Analysis

Multivariate Analysis

Through analysis of the IBM dataset in Milestone 3, we saw that Payment Amount and Payment Format are likely to be important features to predict money laundering due to their strong correlations with the target variable. Specifically, we uncovered that ACH is the primary payment format for fraudulent transactions, followed by check and credit card. Additionally, large payment amounts are highly correlated with fraudulent transactions. Exploring the multivariate relationship between payment format, amount, and the target, we can see in Figure 1 that over 84% of fraudulent transactions comprise large ACH transactions. Conversely, non-fraudulent ACH transactions have a lower mean payment amount. Other payment formats have a significantly lower mean payment amount and are more consistent across the fraudulent and non-fraudulent groups. We can expect that models can try to classify the target by identifying large ACH transactions and disqualifier transactions with Reinvestment or Wire payment format, since the dataset does not contain any fraudulent transactions for these formats.

For the ECC dataset multivariate analysis I explored the relationship between the most correlated features with the target. In a bubble plot of V10 and V14 colored by the target and

sized by V4 we can see a strong separation of the target (see Figure 2). As a result, we can expect that these features could be important in models. Notably, there is still some overlap between the target classes at the edges of the grouping. Non-linear modeling may be required to further distinguish the target. Next, I trained a Decision Tree Classifier model using default parameters to uncover if these features comprise the top important features. In the model feature importance, V14 dominates with a gini importance of 0.76, followed by V17 with 0.07, V10 with 0.02, and V4 is the 6th most important feature with 0.01 gini importance. V14 could be capturing some of the same variance in the target as the other features, resulting in lower gini importance for those features despite strong correlations. Additionally, the model had strong performance on the test set with a f1-score of 0.98 on the negative class and 0.83 on the positive class. The PCA transformation applied to the features in preprocessing could have removed noise, providing a strong result with a basic model and no parameter tuning or class imbalance handling.

In the PPP dataset Milestone 3 analysis, we found `ForgivenDate`, `ForgivenPercentage`, `LoanStatus`, and `LoanStatusDate`, had data leakage of the target, resulting in disqualification. Since no strong correlations with the target remained after disqualification, models need to rely on less strong correlations to predict the target. For the multivariate analysis, I explored whether location and loan application features could provide insight into how the target can be separated. In a bubble plot of the loan `Term`, `TotalProceed`, `CurrentApprovalAmount`, colored by the target, we can see a separation of the majority of the target. We can see that the `Term` length is the main contributor. Fraudulent loans appear between 0-10 or 40-55 months, whereas valid loans are around 25 or 60 months. Additionally, the other features do not distinguish the target in the main groupings, but could still be useful for non-linear patterns. When evaluating models built on this

dataset, I expect to see Term as a strong predictor of the target and the TotalProceed and CurrentApprovalAmount should be less important.

Random Forest Model

The IBM dataset contained over 5 million instances, with 5,177 instances of the positive class (1,000:1 positive negative ratio). I first created a train test split and trained a Random Forest Classifier for baseline model results. Evaluated against the test set, the model had a perfect precision and recall for the negative class, but only 0.13 precision and 0.08 recall for the positive class. The Random Forest did not generalize well to the positive class likely due to the class imbalance. I then undersampled the majority class to bring the dataset to a 10:1 ratio with 56,947 total instances. Next, I tuned the `n_estimators`, `max_depth`, and `min_samples_split` parameters. I executed 100 trials of parameter tuning with cross validation, optimizing for recall and f1. Selecting the best performing parameters and training new models we see a large lift in performance (see Figure 4). The f1 optimized model achieves similar recall to the recall optimized model with increased precision, resulting in less manual effort in reviewing false positives. However, if identifying the maximum number of fraudulent transactions is essential, then the first model is a better choice. Looking at the feature importance of the model, Payment_Format_ACH leads with gini importance of 0.33, followed by Amount_Paid_USD 0.12, Amount_Received_USD 0.11, Payment_Format_Cheque 0.09, and Payment_Format_Credit_Card 0.06. The important features Payment_Format_ACH and Amount_Paid_USD align well with the hypothesis that large ACH transactions could be a splitting criterion. Additionally, the Cheque and Credit Card formats being the next most important aligns with Figure 1, where we see the count of these payment formats is the highest after ACH, making them important for the model to separate the highest number of instances.

The ECC dataset includes over 284,000 instances with 492 positive class instances (577:1 ratio). I utilized the same train/test split and baseline modeling approach as in the previous analysis. The baseline model had strong performance with default parameters and no re-sampling. The negative class attains a perfect precision and recall on the test set and the positive class 0.94 precision and 0.82 recall respectively, resulting in 0.94 macro average f1. To further increase the performance, I leveraged undersampling to create a 5,412 instance dataset and performed parameter tuning. This resulted in lift of macro average f1 to 0.96 with a slight decrease in negative class performance and a lift in positive class (see Figure 5). I hypothesize that the increased baseline performance could be due to less class imbalance and the noise reduction through PCA transformation, as opposed to the IBM dataset. The most important features to the models according to their gini importance were V14 with 0.26, V4 0.11, V10 0.09, V17 0.09, V12 0.09, V16 0.08, and V3 0.08. V14 and V10 comprising two of the important features align with the separation of the target found in Figure 2.

The PPP dataset includes over 968,000 instances with 95 instances of fraud (10,000:1 ratio). By performing the same experiment as the previous datasets, I saw a baseline performance of perfect precision and recall on the negative class and 0.00 precision/recall on the positive class. I addressed the class imbalance by undersampling the negative class and utilized the same parameter tuning techniques. Evaluating the resulting recall and f1 optimized models, we see a large lift in performance. Both models maintain strong performance for the negative class and the positive class f1 increases for each model from 0.00 to 0.45 and 0.81 respectively (see Figure 6). Of the two models, the f1 optimized model performs better across both precision and recall of the negative and positive class, likely due to the small dataset size providing inconsistent results. The model had the following gini importance: Term 0.17, PROCEED_per_job 0.13,

ProcessingMethod 0.10, UTILITIES_PROCEED 0.09, and BusinessAgeDescription 0.08.

Shown in Figure 3, Term stands out as an important feature and is reflected in the model feature importance. Additionally, since the other features rank so highly despite the bubble plot showing that they did not separate the target, the model may be utilizing non-linear patterns.

Comparing Weeks 1-7 to 8-12

From the IBM dataset univariate analysis the Amount Paid feature was shown to be skewed and we hypothesized it could affect model performance. Due to the Random Forest model selection, the model still generalized without log transformations. Additionally, class imbalance affected the positive class model predictions. After undersampling the negative class, the positive class performance improved with maintained negative class performance (see Figure 4). This reinforced that class imbalance was an issue in the dataset. In the bivariate analysis, we showed correlation between payment amounts and formats. We expanded this relationship by demonstrating that the payment amount, format, and target analyzed together show a relationship that can be utilized to separate the target. The feature importance reinforced this idea, with the ACH payment format and payment amount in the top ranking important features.

In the ECC univariate analysis, we found the majority of features were PCA transformed except for Amount and Timestamp. I hypothesized that the Amount feature's right skew would affect the model performance. However, in the modeling phase, Amount was not considered an important feature. Next, I hypothesized class imbalance would be an issue. However, the baseline model received strong performance without parameter tuning or complex model selection. This could be due to the lesser relative imbalance or the strong PCA transformed feature correlations. Finally, in the bivariate analysis I found a number of features with strong correlations. In the feature importance, many of these features were utilized by the model, such

as V14, V17, and V10. The V14 feature's gini importance of 0.76 shows it was able to capture the majority of the variance in the target, despite V4 and V11 also having strong correlations.

In the univariate analysis of the PPP dataset, the dataset contained many right skewed features, which could affect model performance. Despite skew, the model achieved strong performance. Additionally, class imbalance was observed, which affected the model results. I observed in parameter tuning that the results were inconsistent depending on the cross validation folds. As shown in Figure 6 where the second model generalized better, model generalization consistency was an issue. In the bivariate analysis, all the strongly correlated features with the target had to be disqualified due to data leakage. As a result, in the multivariate analysis we were limited to exploring features with less strong correlations. Although we successfully found a set of features which could separate the target, the strongest separator, loan Term, did not seem like a strong initial candidate for predicting fraud (see Figure 3).

Data Conclusions

Expected/Unexpected Data

The most expected aspect of the IBM dataset were the important features found in the model. In Milestone 3, I surmised that Payment Amount, Payment Format, Receiving Amount, and Receiving Format are likely to be important features in models to predict money laundering. Through the multivariate analysis we saw that ACH format and transaction amount were key identifiers of fraudulent transactions. An unexpected finding was that when evaluating a baseline Random Forest model on the dataset without re-sampling the dataset, setting the class_weight parameter to balanced decreased the positive class performance. This could have been due to the splitting criteria not being enough to overpower the large class imbalance.

The ECC dataset baseline model had unexpectedly strong model performance compared to the other datasets despite class imbalance. I hypothesize that this was due to the majority of features in the dataset being PCA transformed in the author's preprocessing, resulting in the model being able to find the patterns in the dataset. Additionally, the class imbalance ratio was significantly lower than the other two datasets, which could have contributed to the improved baseline performance. Additionally, I was surprised to see that the transaction Amount feature was not an important feature to the model, unlike the IBM and PPP dataset. Although the other features are PCA transformed and anonymized as a result, it would have been interesting to understand the underlying features that were important to the model. These features could have been more specific account related features, such as the account's average transaction amount, which provides more context to each transaction relative to the user's regular behaviors.

In the PPP dataset, the target variable positive class size was unexpected. The original dataset includes a separate file for the labels and data, which can be joined on business name during EDA. When joining, I discovered that only 83 of the 100 fraud cases were available in the dataset. Had the join been more successful, there could have been less issues with generalization and overfitting in the modeling phase. Additionally, performance could have been lifted by joining more supporting datasets to include additional features such as tax and business records.

Proposal for Capstone in Module C

In the IBM dataset, we were able to achieve a strong recall of the positive class using the Random Forest model. To improve model performance further, we can increase the precision to reduce the number of false positives. Gradient Boosting models could lead to an increase in precision while retaining strong recall by sequentially building trees to correct the errors. To evaluate the models, I recommend using the same stratified cross validation methodology with

parameter tuning to select models based on f1 and recall metrics that I demonstrated in this paper. Analyzing both Random Forest and Boosting will offer clear insights into how iterative and parallel tree building differs in relation to fraud detection. This analysis can be used to recommend approaches to fraud detection in businesses.

Through the Random Forest model on the ECC dataset, we saw strong results on both the positive and negative class. Due to the performance achieved, I recommend evaluating simpler model types to determine whether simpler models can perform similarly to the ensemble model we utilized in this paper. By using simpler models such as Decision Tree classifier, we can create a more explainable model which can be connected to the underlying data. This would allow businesses to have greater insight into the rules the model relies on for detecting fraud.

For the PPP dataset, a Bagging Classifier model should be evaluated to reduce the overfitting due to the small positive class size. Although the Random Forest should decrease variance by adding randomization to feature selection, evaluating the model would allow performance comparison to validate the assumption. Additionally, recall and f1 should be used in tandem with Leave-one-out cross validation, due to the small positive class size. Class imbalance is a common issue in the industry, and the PPP dataset takes the issue to an extreme. Successful implementation of models on this dataset will reveal reusable insights for other datasets.

Teamwork Reflection

Throughout the semester each of my teammates completed all required work on FeedbackFruits. Rebecca Choi, Kyle Moise, and Paramdeep Nijjer provided timeline and thoughtful feedback on my assignments across each of the feedback categories. Our group's agreed-upon collaboration did not extend beyond FeedbackFruits, but we did create a group chat to have open discussion if anyone had issues throughout the assignments.

Figure 1: IBM dataset Heatmap of Amount Paid by Payment Format and Target

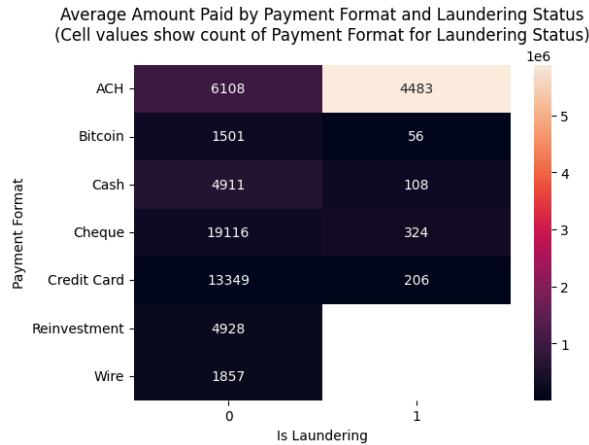


Figure 2: ECC dataset bubble plot of strongest correlations colored by target

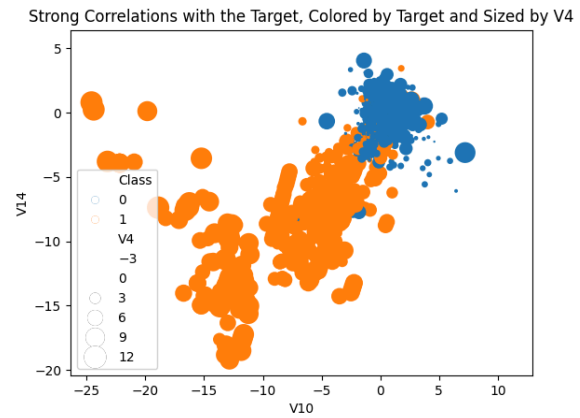


Figure 3: PPP dataset bubble plot of features colored by target

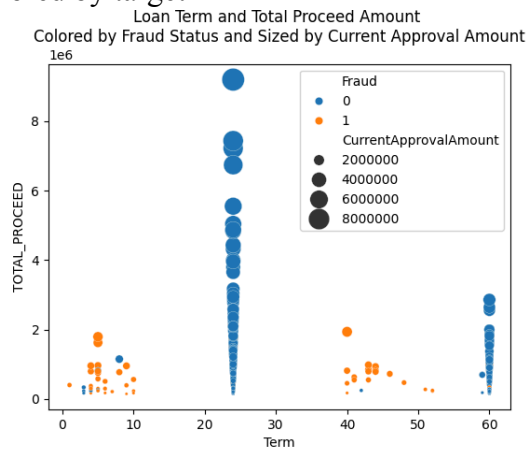


Figure 4: IBM dataset classification report

		precision	recall	f1-score	support
IBM RF optimized for Recall	0	0.983030	0.928634	0.955058	10355.000000
	1	0.540423	0.839614	0.657586	1035.000000
	accuracy	0.920544	0.920544	0.920544	0.920544
	macro avg	0.761726	0.884124	0.806322	11390.000000
	weighted avg	0.942811	0.920544	0.928027	11390.000000
IBM RF optimized for F1	0	0.978766	0.948141	0.963210	10355.000000
	1	0.604857	0.794203	0.686717	1035.000000
	accuracy	0.934153	0.934153	0.934153	0.934153
	macro avg	0.791811	0.871172	0.824963	11390.000000
	weighted avg	0.944789	0.934153	0.938085	11390.000000

Figure 5: ECC dataset classification report

		precision	recall	f1-score	support
CC RF optimized for Recall	0	0.990891	0.993909	0.992397	985.000000
	1	0.936842	0.908163	0.922280	98.000000
	accuracy	0.986150	0.986150	0.986150	0.986150
	macro avg	0.963866	0.951036	0.957339	1083.000000
	weighted avg	0.986000	0.986150	0.986052	1083.000000
CC RF optimized for F1	0	0.989930	0.997970	0.993933	985.000000
	1	0.977778	0.897959	0.936170	98.000000
	accuracy	0.988920	0.988920	0.988920	0.988920
	macro avg	0.983854	0.947964	0.965052	1083.000000
	weighted avg	0.988830	0.988920	0.988706	1083.000000

Figure 6: PPP dataset classification report

		precision	recall	f1-score	support
PPP RF optimized for Recall	0	0.969880	0.847368	0.904494	190.000000
	1	0.325581	0.736842	0.451613	19.000000
	accuracy	0.837321	0.837321	0.837321	0.837321
	macro avg	0.647730	0.792105	0.678054	209.000000
	weighted avg	0.911307	0.837321	0.863323	209.000000
PPP RF optimized for F1	0	0.979058	0.984211	0.981627	190.000000
	1	0.833333	0.789474	0.810811	19.000000
	accuracy	0.966507	0.966507	0.966507	0.966507
	macro avg	0.906195	0.886842	0.896219	209.000000
	weighted avg	0.965810	0.966507	0.966099	209.000000

References

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AI Appendix

I did not use AI for this submission.